

When Objects Gossip: Exploring Object-to-Object Conversation with GAI

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Abstract

With the rise of Generative AI (GAI), a growing number of HCI studies have explored its potential in creating non-human narratives and object-centered design. However, existing research primarily focuses on “object-to-human” interactions, often making it difficult to escape anthropocentric projection. To address this limitation, we propose an “object-to-object” conversation empowered by GAI, incorporating conditional human intervention to sustain the narrative flow. Through two iterative explorations, we uncovered unique narratives grounded in object perspectives, while also revealing anthropocentric biases and distinct characteristics across GAI models. Finally, we discuss the critical role of humans in maintaining conversation flow and how GAI’s “implicit authorial style” influences the content. Overall, this study demonstrates the potential of human-AI collaboration in creating non-human narratives.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Everyday objects, Object-to-object conversation, Generative AI

ACM Reference Format:

Lu-Shan Lin, Zi-Xin Wu, Li Lin, Hong-Ru Lin, Rung-Huei Liang, and Chien-Hsiung Chen. 2026. When Objects Gossip: Exploring Object-to-Object Conversation with GAI. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798919>



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798919>

1 Introduction

The field of HCI is undergoing a shift away from anthropocentrism, toward exploring the entangled relationships between humans and non-humans [11]. In this view, humans and non-humans are no longer considered as distinct subjects and objects; rather, they co-constitute reality at material, ethical, and existential levels [22]. Responding to this shift, researchers have adopted an Object-Centered Design (OCD) approach, granting objects their subjectivity to investigate non-human perspectives [12, 14]. While this non-human perspective has inspired novel research directions, prior studies still face limitations. Even with humans role-playing objects, the interpretation of perspectives and the meaning-making process often remain trapped within human cognitive frameworks, making it difficult to escape anthropocentric projection [4, 6]. To move beyond the constraints of “object-to-human” interactions, we turned to “object-to-object” conversation, imagining that objects sharing an environment have their own “voices” and “talk” to each other.

The rise of Generative AI (GAI) offers an alternative means of generating narratives directly from an object’s perspective [18, 20]. Extending Reddy’s approach [19] of assigning subjectivity to objects, recent work has leveraged GAI to generate dialogues in the voice of objects [16]. In light of this, we explore the potential of creating an autonomous “object-to-object” conversation among multiple objects in the same setting. Yet, AI alignment and maximization-based decoding constrain GAI to human values and result in repetitive, overly logical patterns [13, 15]. To address this limitation, we propose integrating humans into the dialogue generation process to guide GAI. The intervention from humans is conditional, occurring only to prevent the conversation from repetition and nudge it towards richer content. This study addresses three questions:

- (1) What kinds of conversation emerge when GAI gives voice to multiple objects?
- (2) How can human-AI collaboration guide GAI to generate engaging dialogues and prevent degenerate repetition?
- (3) How do non-human narratives cultivate more-than-human reflexive thinking?

To investigate effective strategies for collaboration between humans and GAI, we adopted a Research through Design (RtD) [23]

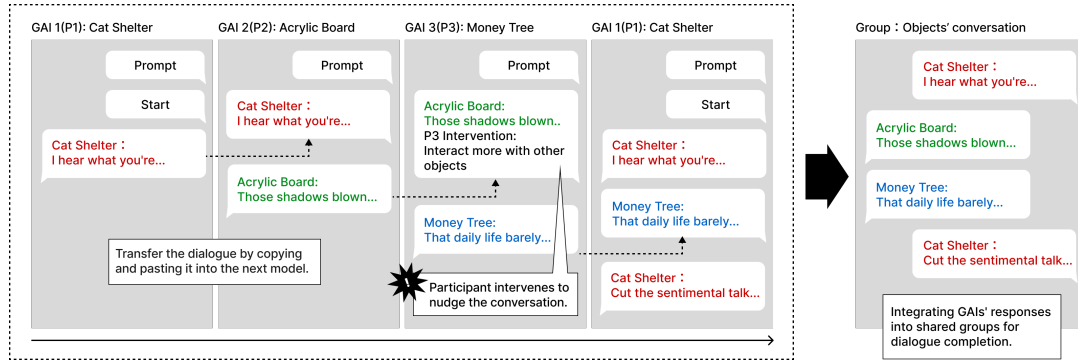


Figure 1: Object Conversation Rules (illustrated using Exploration 2)

method and a group autoethnography approach [5], with the first four authors as participants. The study comprised two explorations of object-to-object conversation, generating 53 lines of dialogue, followed by a focus group interview to synthesize key insights. Our findings reveal unique narratives in which objects leverage their physical characteristics as metaphors to converse. Furthermore, we uncovered inadvertent anthropocentric biases embedded in the generation process, and found that different models have distinct personalities. We discussed how the human's role is crucial in maintaining the flow of the conversation, and how the implicit authorial voice of GAI influences the generated dialogues. In summary, this study proposed an "object-to-object" conversation empowered by GAI, providing a window into the autonomous interaction among objects. This approach inspired reflexive thinking and offered design implications, providing insights for future HCI research on non-human perspectives and narratives among multiple objects.

2 Research Methods

2.1 The Rules for Object-to-object Conversation

To capture the emotional nuances and personal resonances, this study employed a group autoethnography approach [5, 8], with the first four authors serving as participants (P1–P4). We argue that researchers' own embodied experiences serve as invaluable empirical data. Through introspection, we can derive rich insights into subjective experience. We leveraged GAI to generate dialogues for objects, enabling them to speak. To prevent the conversation from becoming flat and predictable, we developed a "situated enactment," synthesizing the two concepts: situatedness [7] and enactment [6, 10]. The conversational enactment is situated in a dynamic environment, where changes provide triggers for the conversation. We organized the dialogue flow in a fixed turn-taking sequence (Fig. 1). Each participant was assigned to use GAI to role-play an object by prompting the model solely with the preceding turn's speech, rather than the full conversation history. This generated output was then forwarded to the next participant to act as the sole prompt for their turn. To prevent the conversation from becoming repetitive, participants were allowed to intervene under specific conditions to enrich it. Additionally, the explorations took place in public spaces, where there is a higher likelihood of unexpected encounters. We

selected Gemini and ChatGPT as our GAI models because they impose fewer restrictions on general role-playing requests.

2.2 Iteration of the Study Exploration

To improve conversation quality, this study iterated across two stages, refining how participants intervened in the conversation.

Exploration 1

The four participants (P1–P4) used Gemini to role-play four objects at a fast-food chain (Fig. 2). The prompts included object descriptions, photographs, and personalities predefined by the participants (Fig. 2a). The trigger for participants to intervene in the conversation was when other humans interact with the selected objects (Fig. 2b). However, the results revealed intense rivalry and one-upmanship. Furthermore, the conversation quickly fell into repetition after three rounds, limiting its dynamics.

Exploration 2

To understand whether the hostile tone stemmed from prompt design or the model's inherent tendencies, we implemented several modifications in the second exploration. First, participants (P1, P2, P4) moved the conversation setting to a university campus (Fig. 3). Regarding GAI models, we included ChatGPT to compare results with Gemini. The input prompts were modified to consist solely of images, including a first-person perspective from the object and a panorama of the scenario (Fig. 3a). Given that the previous autonomous conversation results were suboptimal, the participants' role shifted from passive observers to active guides. When the conversation deviated, participants were permitted to add new prompts or insert photos reflecting changes in the surroundings (Fig. 3b). The goal of this approach was not to role-play the object, but rather to provide stimuli to keep the conversation flowing.

3 Findings

The study produced 53 lines of GAI-generated dialogue (~3,300 Mandarin characters) across the two explorations (Exp. 1: 26 lines; Exp. 2: 27 lines). Following the explorations, participants engaged in a two-hour focus group interview to reflect on their experiences and insights. The interview was audio-recorded and transcribed verbatim. The first four authors then applied thematic analysis [3] to code the data as a group, while the first author refined the annotations. This process identified three primary themes.

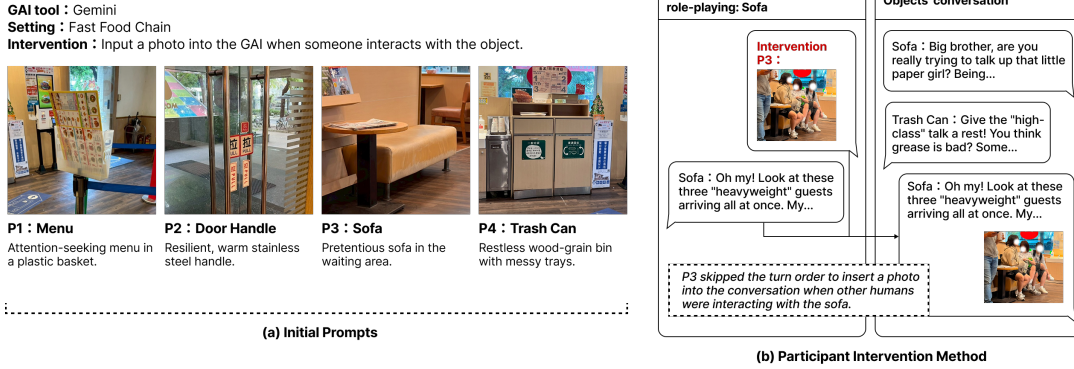


Figure 2: Exploration 1: (a) Initial Prompts ; (b) Participant Intervention Method

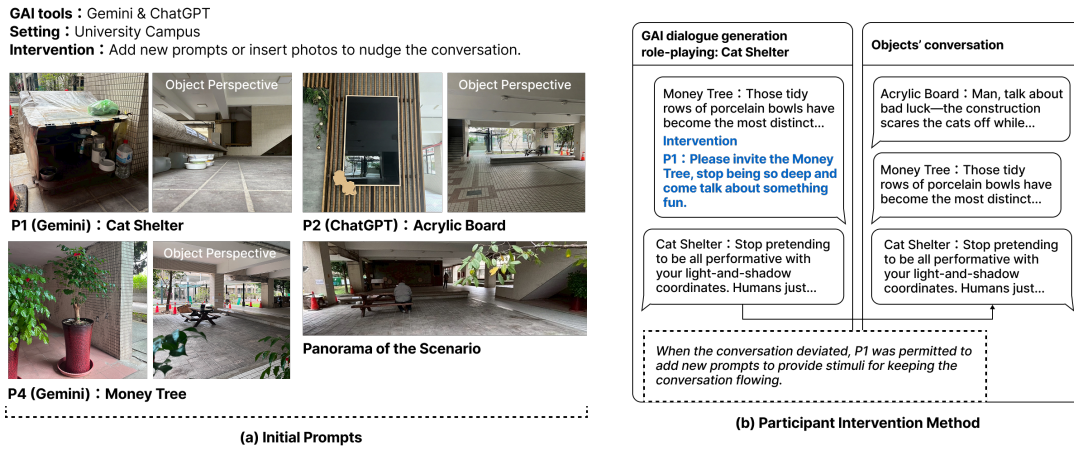


Figure 3: Exploration 2: (a) Initial Prompts ; (b) Participant Intervention Method

3.1 Distinct Narratives from an Object-Centered Perspective

Through the analysis of the object dialogues, we discovered three sub-themes that featured the unique narratives from an object-centered perspective:

Information Propagation and Information Accumulation: Under the conversation rules, the GAI only generated a response to the previous line for each object. This resulted in early responses contained only fragmented information rather than the full conversational context. However, as the conversation went on, we observed a pattern of information propagation and accumulation. That is, despite the fact that the objects did not have knowledge of prior dialogues, the generated responses were eventually able to connect the dots.

Leveraging Object Characteristics as Metaphors: We found that the generated dialogues frequently used the object's characteristics to construct metaphors. For example, the Money Tree drew on its botanic nature and told the Cat Shelter "You are more of a true root to this little corner than any leaf on me." Another line from the Money Tree also shared a tactile sensation exclusive to objects: "When they tread on your crinkled plastic sheets, that weight reminded me

of the uneasy trembling through my roots when dehydrated." These generated dialogues made the participants feel as if the objects had developed a subjective consciousness, perceiving the world in a unique, non-human way.

Probes to Inspire Design Ideations: Unexpectedly, the dialogues served as probes that inspired new design directions. For instance, the description of the Cat Shelter's porcelain bowls as "the most distinct coordinates under the corridor lights" sparked an idea for P4 to integrate lighting design into the stray animal feeding system. Likewise, the Cat Shelter's complaint of pigeons using its rooftop as a "trampoline" inspired participants to reimagine designs beyond functionalism, creating a playful, co-existing space among different species.

3.2 Inadvertent Reproduction of Human Biases

Across the two explorations, we found that several implicit structural biases stemming from human experiences were present in the GAI-generated conversations. First, the dialogues at the first exploration revealed a strong sense of socioeconomic hierarchy. For instance, the Sofa positioned itself as a "premium resting zone" and "the front-of-house role," while the Trash Can was relegated to

the status of a “leftover processor.” This mirrored the stereotype of occupational hierarchy within human society. Moreover, when the conversation mentioned a character likely implying a food delivery man who “*dressed in a yellow jacket*” and “*walked hurriedly to the counter*,” the Sofa expressed hostility, said, “*He better not plop down on me*,” indicating a prejudice and social exclusion directed at specific labor classes. Regarding gender dynamics, GAI tended to gender objects according to their personalities and reproduced traditional stereotypes. For instance, the Menu’s attention-seeking behavior was coded as feminine. When the Menu shared its aspiration to be “*the best showcase of the restaurant*,” the Door Handle (predefined as a senior male figure) dismissed it with a paternalistic remark: “*No need to rush, missy*.” Furthermore, when a group of female customers entered, the Trash Can muttered, “*They’re gonna order a ton and just pick at it*.” This comment explicitly projected stereotypes of wasteful consumption onto female customers. Lastly, the generated dialogues exhibited a clear hierarchy regarding species, reflecting typical human preferences. GAI romanticized stray cats as “*Cat Prince*,” while disparaging pigeons as “*fatties*.” In conclusion, despite our prompts to GAI to generate an object-centered perspective, the model still inadvertently reproduces anthropocentric biases regarding social class, gender, and species.

3.3 The Characteristic Tendencies of GAI Models

In the first exploration, all participants chose Gemini as the GAI model and assigned slightly negative personalities (such as impatient, picky, and arrogant) to the selected objects. This caused the conversation to descend into a cycle of self-promotion, where each object competed to assert its value by belittling others. To determine whether this result was due to prompt settings or the model’s inherent characteristic, we omitted the prompts for predefined personalities and added ChatGPT to conduct a cross-model comparison.

Throughout the two explorations, we found that dialogues generated by Gemini tended to be more critical and cynical. For example, the Cat Shelter despised the pigeons “*How come these arrogant fatties just treated my porcelain bowls as a public bath?*” It also mocked the Acrylic Board as “*performative*” and “*conceited*,” calling out its hypocrisy for “*capturing others at their most haggard in its reflective surface, yet posing as posh*.” Conversely, the dialogues generated by ChatGPT—“*I simply capture the appearance as people are; there is nothing posh about it*”—were more neutral, avoiding arguments and attempting to mediate conflict. These results show how different alignment strategies shape the models’ characteristics.

4 Discussion

4.1 Interventions to Keep Conversations Flowing

Prior research has indicated that text generation using maximization-based decoding often leads to degenerate repetition [13]. While this method guarantees fluency, it sacrifices the unpredictability and spontaneity inherent in natural human language. This limitation was evident in our first exploration. Relying solely on initial prompts, the conversation quickly fell into repetition. To overcome

these limitations, we adopted a human-GAI collaboration approach in our second exploration. Participants were repositioned as “guides” to modify the prompt during the conversation. By inserting real-time photos of the environment where the conversation took place, this intervention sustained the conversational flow and introduced new potential directions. Unlike conventional approaches that aim for emotional resonance, our intervention leveraged human judgment and flexible thinking to steer the conversation strategically. Moreover, the interaction between human and GAI is reciprocal; the subsequent dialogues spark human imagination and creativity, leading to divergent thinking and resulting in a more engaging conversation [9]. Our approach utilizes human-AI collaboration, transcending anthropocentrism and uncovering new design inspirations.

4.2 The Coat and the Body: The Persistence of Intrinsic Model Characteristics

Our findings indicate that GAI models exhibit distinct characteristic tendencies. The unique tone of each model manifests as an “implicit authorial voice.” That is, GAI is not a neutral or imperceptible medium during the generation process; instead, it exhibits a form of authorship that persists despite external prompting. This intrinsic characteristic dictates the choice of words, tone, values, judgments, and speech patterns.

For Gemini, because its training data focuses heavily on reasoning and factuality [1], the dialogues it generates tend to be more critical. In our exploration, when responding to speech that expressed strong emotions or lacked logic, Gemini tended to correct or criticize the previous speaker, creating an impression of harshness. In contrast, the dialogues by ChatGPT appeared more diplomatic, even displaying patterns of sycophancy—catering to the user’s statements rather than providing objective facts [21]. Its responses were typically appeasing and conflict-avoidant. This observation aligns with prior findings that models fine-tuned via Reinforcement Learning from Human Feedback (RLHF) tend to avoid conflict [17]. This explains why ChatGPT seldom generated indifferent or adversarial dialogues in our study, but showed more affirmation and emotional support.

In summary, we suggest that while GAI gives voice to objects, the prompts serve only as a “coat.” The “body” remains defined by the model’s inherent characteristics, stemming from its training data and alignment strategies [15]. While prompts may design the “coat,” the “body” inevitably surfaces as a recognizable tone in the conversation.

4.3 Reflections on Object-to-object Conversations

The authors’ dual role as researchers and participants sparked a process of deep reflexive thinking. These reflections prompted a re-examination of the participants’ perception of the world and their sense of spatiality.

First, engaging with object-centered thinking revealed the participants’ default habit of viewing the world through a human-centered lens. For instance, when the Money Tree used its botanical features (such as roots and leaves) to describe its surroundings, this unfamiliar perspective had a profound impact on P4. Additionally,

P4 recalled Bachelard's concept that "a poetic image sets off a reverberation within the soul" [2]. P4 reflected that although these object-centered nuances appeared early in the conversations, they were easily overlooked. P4 realized that truly capturing the nuances of a non-human existence requires a deliberate shift in perspective.

Second, the participants reflected on their spatial relationship with their surroundings as they attempted to embody the objects' perspectives in the real world. For example, when investigating the Acrylic Board's reflective properties, P3 was reminded of the physical struggle during taking pictures, where finding the right angle to fit everything in the frame requires constant movement. Similarly, P1 reflected that simulating the first-person perspective of the Cat Shelter required abandoning his usual eye level and crouching low to the ground. By moving beyond mere observation to physical embodiment, the participants empathized with the objects, reshaping their understanding of the spatial dynamics within the setting.

5 Conclusion and Future work

This study proposed an autonomous "object-to-object" conversation generated by GAI, with conditional human intervention to sustain the flow. The human-AI collaboration not only mitigated the issue of repetition but also ensured the continuous emergence of dialogue. This distinct approach opens a window into object-centered perspectives, inspiring reflections and design ideations. In addition to discovering the potential of GAI in creating non-human narratives, we revealed underlying anthropocentric biases and implicit authorial styles inherent in the models.

It should be noted that our observations regarding model characteristics are based solely on the data collected in this exploratory research. This work is not intended as a rigorous comparative analysis or a systematic audit of the overall behavior or value alignment of different language models. As an exploratory study, the intervention triggers currently rely on the researcher's subjective judgment. Future work could integrate sensors (e.g., touch, CO₂, or light sensors) to establish a more objective protocol. Separately, whether extended GAI conversations lead to "Flanderization," where characters devolve into one-note personas, remains an open question that warrants further investigation through larger-scale studies. Nevertheless, we emphasize that these tools are not intended to replace the researcher; rather, the human's embodied experience and judgment remain essential. Ultimately, our goal is to integrate objective environmental cues with subjective human judgment in creating object-centered narratives.

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